# Practice workbook · Show how an image forms on the retina and reaches the brain

Continuing rays through cornea and lens to an inverted image on the retina, and the optic nerve carrying impulses to the brain, including effects of damage.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EYE-IMAGE-FORMATION-AND-OPTIC-NERVE

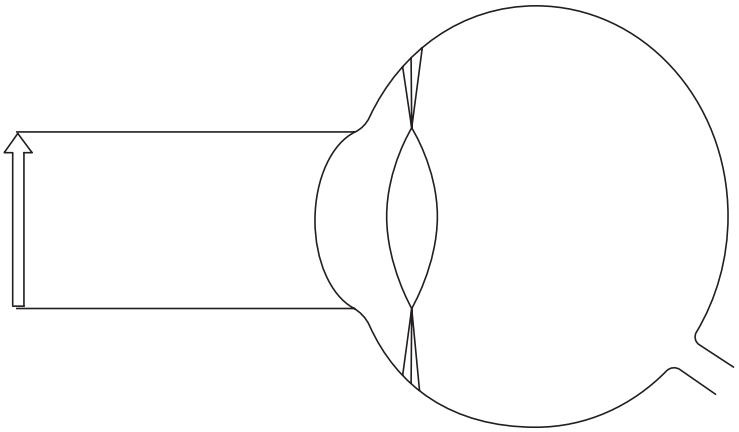
# 1. Question 8 · part a.i · 5 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.i only.

**8** The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

**(5)**

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

**(2)**

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

**(1)**

(ii) Give a feature of the eye that can lead to long sight.

**(1)**

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## 1. Question 8 · part a.i · 5 marks · Working and answer

Working and answer

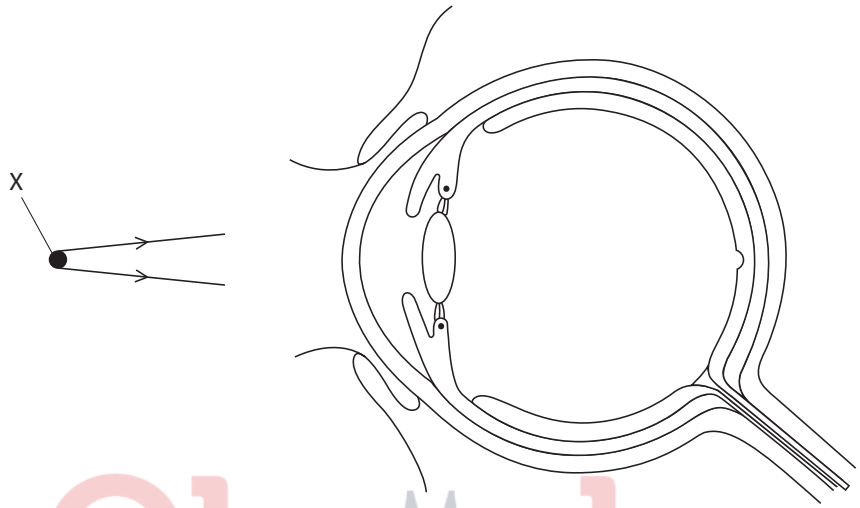
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## 2. Question 1 · part b · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

**1** The diagram shows a section through an eye.




(a) The table lists descriptions of parts of the eye.

Use the letters, A, B, C, D and E, to label the parts of the eye on the diagram.

**(5)**

| Description of part | Letter |
|---|---|
| the layer that contains light sensitive cells | A |
| the muscles that change lens shape | B |
| the structure that controls the amount of light entering the eye | C |
| the structure that carries nerve impulses away from the eye | D |
| the structure that protects the eye from dust and grit | E |

(b) A person looks at object X.

Complete the diagram by continuing the rays of light to show how the image of object X is formed on the retina.

**(3)**

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## 2. Question 1 · part b · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

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(c) Name three defects of the eye that can be corrected by using glasses.

**(3)**

1 .....

2 .....

3 .....

**(Total for Question 1 = 11 marks)**

Working and answer

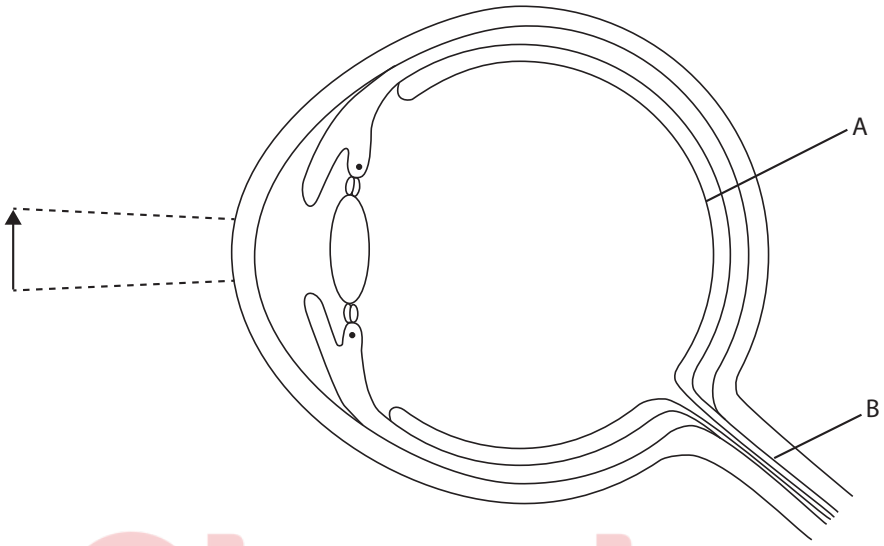
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## 3. Question 6 · part a.ii · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.ii only.

**6** The diagram shows a section through the eye.




(a) (i) Describe the functions of structures A and B. **(4)**

A .....

.....

.....

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B .....

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(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

**(4)**

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P 7 8 7 6 9 A 0 1 4 2 0

## 3. Question 6 · part a.ii · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.ii only.

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(b) Describe a simple experiment to show that stereoscopic vision would help a person to catch a ball thrown towards them.

**(4)**

(c) Describe what is meant by the term **astigmatism**.

**(2)**

**(Total for Question 6 = 14 marks)**

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## 3. Question 6 · part a.ii · 4 marks · Working and answer

Working and answer

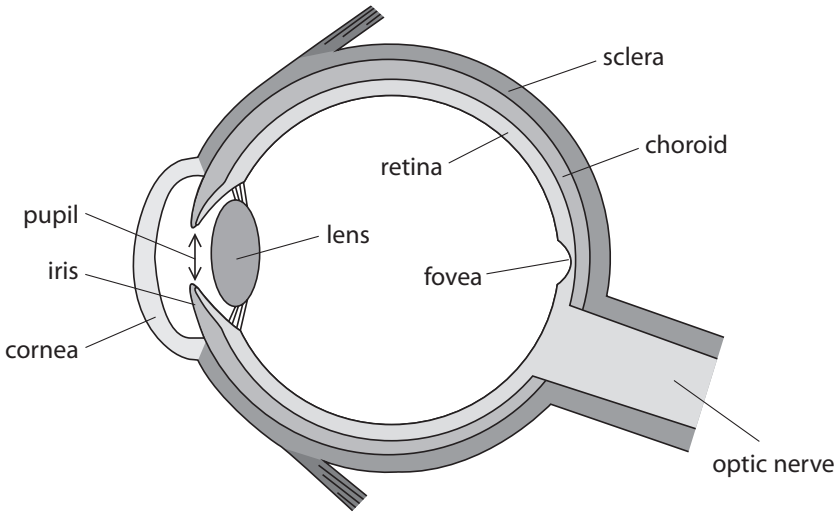
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# 4. Question 3 · part c.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.i only.

**3** (a) The diagram shows the human eye, with some structures labelled.




(i) Which structure helps to focus light at the back of the eye? **(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye. **(3)**

| | | | | |
|---|---|---|---|---|
| sclera | retina | choroid | iris | pupil |

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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# 4. Question 3 · part c.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.i only.

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(b) The photomicrograph gives the thickness of the two outer layers of the eye.




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Estimate the thickness of the retina.

**(1)**

thickness of the retina = ..... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

**(2)**

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(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

**(2)**

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**(Total for Question 3 = 9 marks)**

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## 4. Question 3 · part c.i · 2 marks · Working and answer

Working and answer

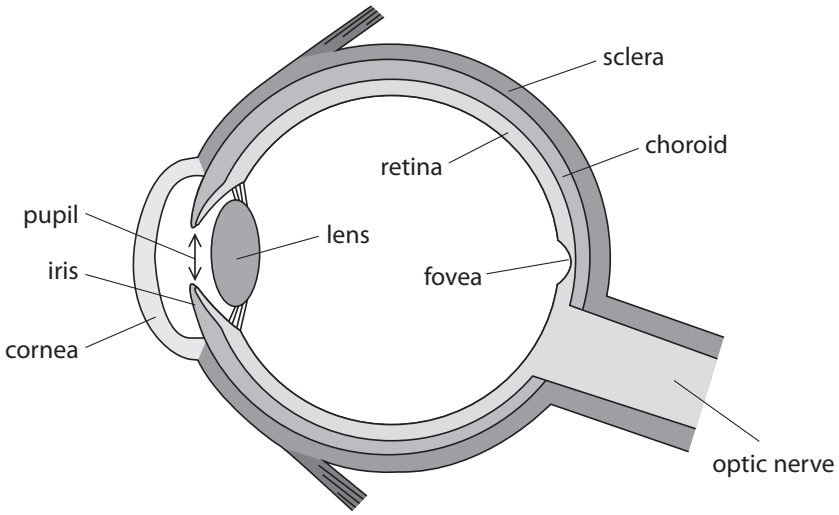
OLevels.com SCHOOL

# 5. Question 3 · part c.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.ii only.

**3** (a) The diagram shows the human eye, with some structures labelled.




(i) Which structure helps to focus light at the back of the eye?

**(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye.

**(3)**

| | | | | |
|---|---|---|---|---|
| sclera | retina | choroid | iris | pupil |

Use words from the box to complete the passage about the eye.

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# 5. Question 3 · part c.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.ii only.

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(b) The photomicrograph gives the thickness of the two outer layers of the eye.




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Estimate the thickness of the retina.

**(1)**

thickness of the retina = ..... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

**(2)**

.....

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(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

**(2)**

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**(Total for Question 3 = 9 marks)**

## 5. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

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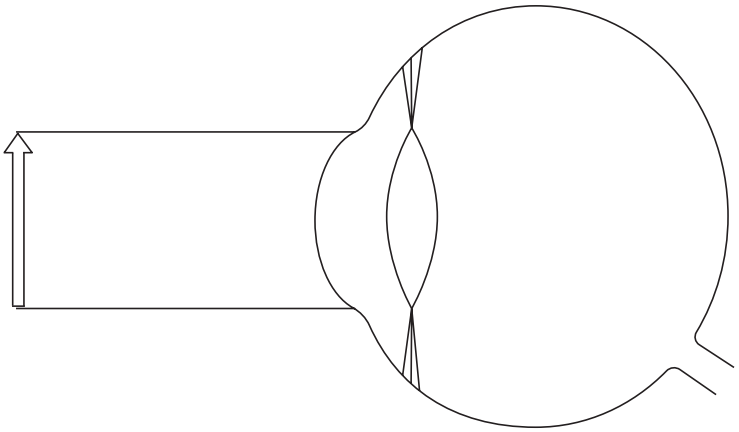
## 6. Question 8 · part a.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

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**(5)**

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

**(2)**

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(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

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## 6. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

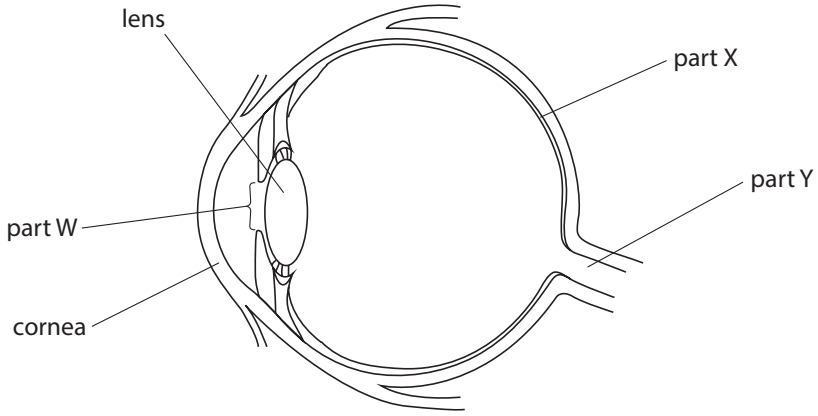
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# 7. Question 3 · part a.ii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.ii only.

**3** (a) The diagram shows the human eye.




(i) Name part W and part X.

**(2)**

W .....

X .....

(ii) Describe the function of part Y.

**(2)**

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

**(2)**

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## 7. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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